

Personalised Learning Engagement And Recommendation System

QUIZ (10 MCQs)

Q1. What was the primary goal of the student project?

- A. Predict student income
- B. Predict student academic outcomes
- C. Rank students
- D. Build a dashboard

Correct Answer: B

Explanation: The model predicts academic performance or outcomes.

Q2. Why was exploratory data analysis important in this project?

- A. To remove all features
- B. To visualize student behavior patterns
- C. To improve hardware performance
- D. To tune hyperparameters

Correct Answer: B

Explanation: EDA helps understand patterns and data quality issues.

Q3. Why were categorical features encoded before training?

- A. Models cannot directly handle text data
- B. Encoding increases dataset size
- C. Encoding improves accuracy always
- D. Encoding is optional

Correct Answer: A

Explanation: Most ML models require numerical inputs.

Q4. What does the target variable represent in this project?

- A. Student ID
- B. Teacher rating
- C. Student outcome or performance label
- D. Attendance count

Correct Answer: C

Explanation: The target reflects academic performance or result.

Q5. Why is train–test split necessary?

- A. To increase data
- B. To reduce noise
- C. To evaluate model performance on unseen data
- D. To balance classes

Correct Answer: C

Explanation: It helps assess generalization.

Q6. Why did we evaluate multiple models?

- A. To increase training time
- B. To compare performance and behavior
- C. To satisfy course requirements
- D. To reduce preprocessing

Correct Answer: B

Explanation: Different models capture patterns differently.

Q7. What does feature importance tell us?

- A. Model accuracy
- B. Which features influence predictions most
- C. Data leakage
- D. Overfitting level

Correct Answer: B

Explanation: It shows relative influence of input features.

Q8. Why is overfitting a concern in student prediction models?

- A. Data size is small
- B. Predictions may not generalize to new students
- C. Overfitting increases accuracy
- D. It slows training

Correct Answer: B

Explanation: Overfit models fail on unseen students.

Q9. How can missing student data affect predictions?

- A. It has no effect
- B. It improves accuracy
- C. It can introduce bias or incorrect predictions
- D. It removes features

Correct Answer: C

Explanation: Missing data alters feature distributions.

Q10. Why is interpretability important in this project?

- A. To impress users
- B. To understand why students succeed or struggle
- C. To increase speed
- D. To reduce preprocessing

Correct Answer: B

Explanation: Educators need understandable insights.
